SKOPJE · CIRCULAR ECONOMY PLAN

Circular Economy for Skopje

Seven steps that turn the city's waste into recycled materials, electricity and heat for our own homes.

AI sorting bins | Autonomous trucks | Humanoid robots | Digital twins | Waste-to-energy

A concept presentation · 2026 · Photo: Pudelek, CC BY-SA 4.0 (Wikimedia Commons)

THE STARTING POINT

Why Skopje needs a circular economy

Today the city pays twice for every kilo of waste: once to buy the material, and again to bury it.



01

Burying waste is the most expensive option

Landfill space is limited, the site keeps growing, and every tonne buried is paid for twice.

02

Our air is polluted

Skopje is regularly ranked among Europe's most polluted cities. Open waste burning and wood and coal heating are a big part of it.

03

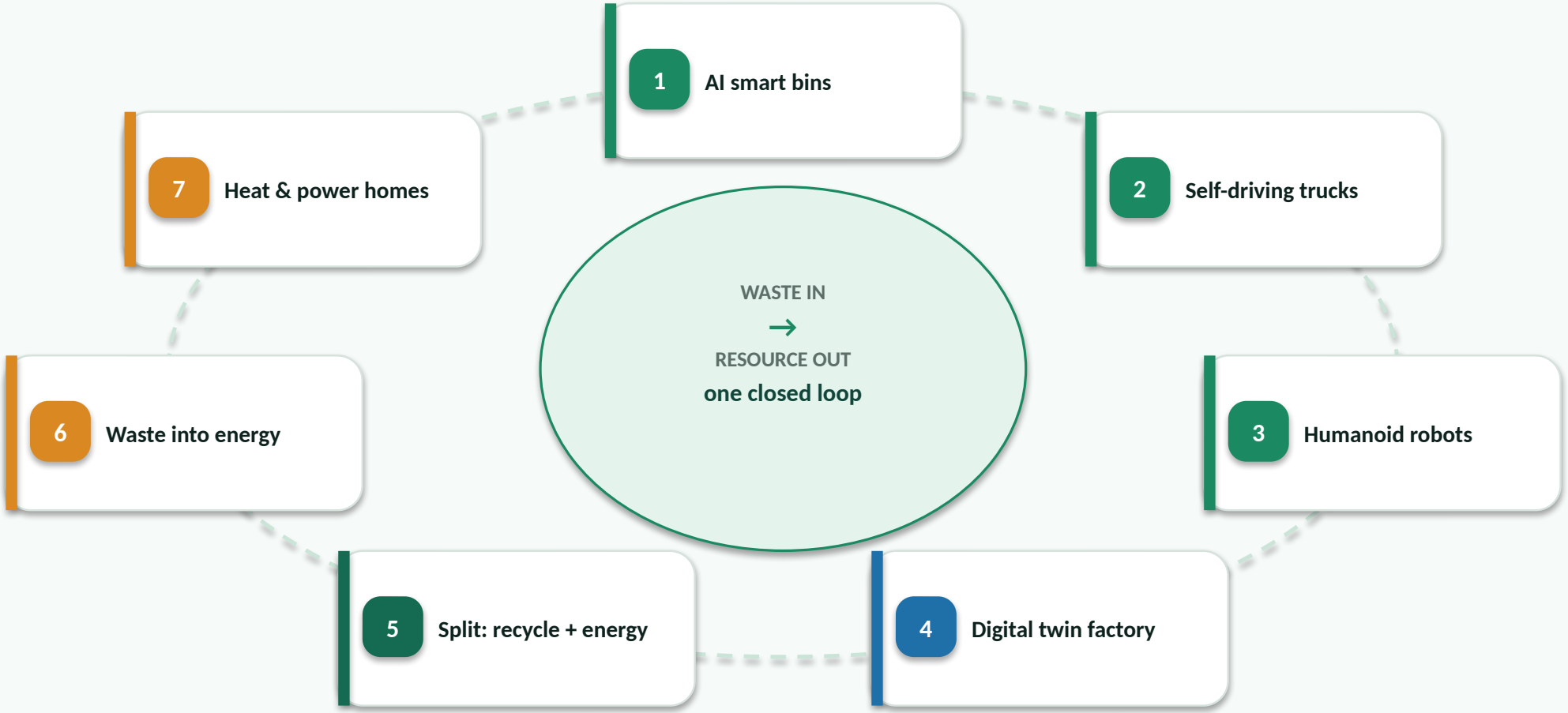
We throw away value — and energy security

Paper, plastic, glass, metal and food are raw materials we already paid for. And we import much of the fuel we heat with.

Skopje does not have a waste problem. It has a resource it is not using yet.

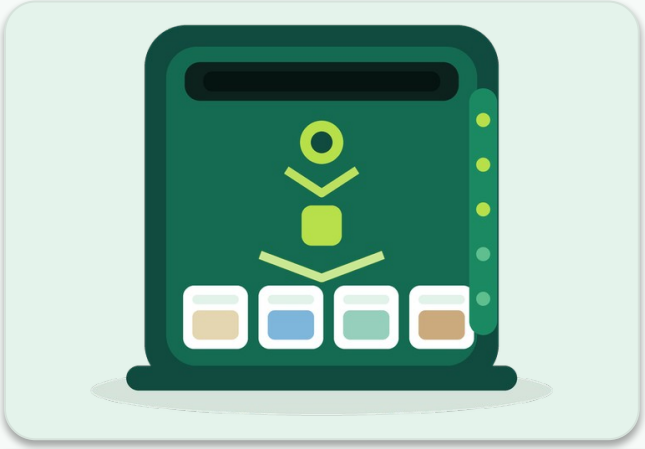
THE WORKFLOW

The 7 steps of Skopje's circular economy



STEPS 1 & 2 · COLLECTION AND TRANSPORT

Sorted at the kerb, moved by robot

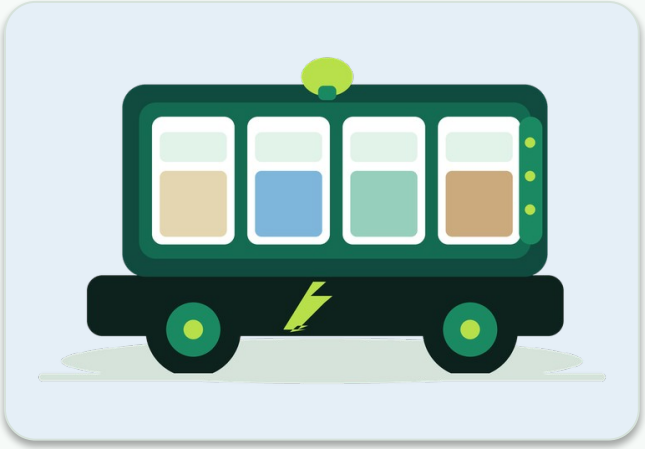


STEP 1 · COLLECT

Bin-e, Poland

AI bins sort the waste for you

You drop everything into one opening. A camera and an AI model recognise the material and send it to the right compartment — about 95% accurate. Paper and plastic are compressed, and fill-level sensors tell the city when a bin is full, so trucks only drive when needed.



STEP 2 · TRANSPORT

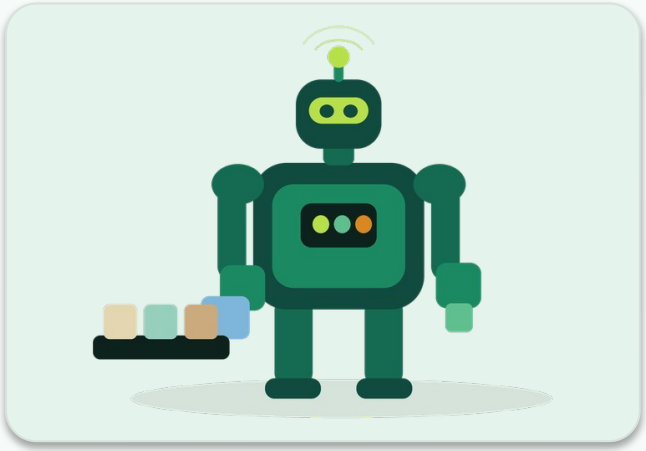
Einride, Sweden

Self-driving electric trucks move it

Cabless, fully electric, Level 4 autonomous trucks arrive quietly at night and empty the bins with robotic arms. Each truck is a moving warehouse with a sealed chamber for every waste type — paper never touches glass. Driverless electric freight already runs daily on public roads in Europe and the US.

STEPS 3 & 4 · AUTOMATION AND INTELLIGENCE

The factory that runs itself

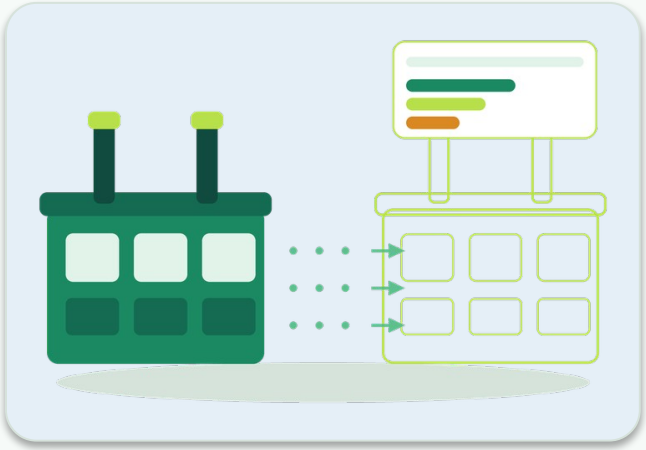


STEP 3 · HANDLE

Neura Robotics, Germany

Humanoid robots run the logistics

Inside the plant the robots unload the trucks, move each fraction to the right line, feed the machines and check quality — the dirty, repetitive, 24/7 work. They see in 360°, sense touch, and work safely beside people. It is also where the new skilled jobs are created.



STEP 4 · OPTIMISE

NVIDIA Omniverse

A digital twin designs and runs the plant

Before any concrete is poured, the whole factory is built and tested as a live 3D simulation fed by real sensor data. Bottlenecks get fixed in the computer instead of in the plant, and the twin keeps optimising it for 24/7 operation. BMW projects around 30% savings from planning this way.

STEPS 5 & 6 · SEPARATION AND CONVERSION

Two streams: materials and energy



Real plant, real technology. MARTIN GmbH of Munich builds the furnaces and grates used in waste-to-energy plants in Vienna, Augsburg and Magdeburg.

STREAM A · RECYCLING

Paper, plastic, glass, metal and electronics are cleaned and sold back to industry as raw material. The city earns from what it used to pay to bury.

STREAM B · ENERGY

Whatever cannot be recycled is burned in a modern controlled plant. The heat makes steam, the steam spins a turbine: electricity for the grid and hot water for district heating.

Recycle first. Recover energy from what is left over. Bury almost nothing.

STEP 7 • CLOSING THE LOOP

What Skopje gets back



- **Lower bills.** Heat and power made locally, from waste we already own instead of fuel we import.
- **Cleaner air.** Fewer wood and coal stoves and less open burning — better air for children and the elderly.
- **Energy security.** A shock in world markets cannot switch off the city's heating.
- **New skilled jobs.** Plant operators, technicians, robotics and data specialists — careers worth staying for.
- **Local investment.** The money stays in Skopje instead of going to landfill operators and foreign fuel suppliers.

What you put in the bin comes back as the warmth in your radiator.

PROOF FROM THE REAL WORLD

Other countries already do this — and gained from it

Not pilot projects or promises. Operating cities, with decades of measured results.



AUSTRIA

Vienna — Spittelau

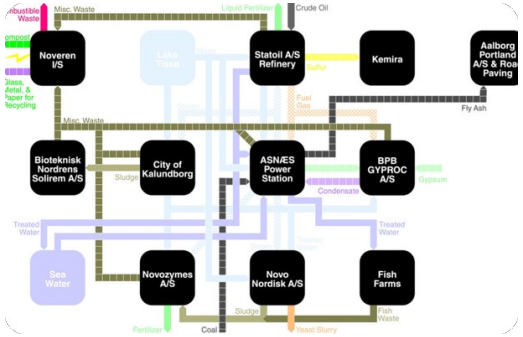
250,000 tonnes of waste a year become heat for 60,000+ households. Runs about 90% below the legal emission limits.



DENMARK

Copenhagen — CopenHill

Heat and power for around 150,000 households — with a public ski slope, trail and climbing wall on the roof.



DENMARK

Kalundborg Symbiosis

17 companies exchanging waste as raw material since 1972. Saves ~635,000 t CO_2 and ~€80 million every year.



JAPAN

Kawasaki Eco-Town

Urban-industrial symbiosis since 1997. Waste sent for incineration fell by 200,000 tonnes while the population grew.

Sweden goes further still: waste-to-energy heats about 1.47 million apartments — and Sweden imports waste to keep those plants running.

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THE EVIDENCE

Real examples, real gains — it already works

Countries that built this system did not just help the planet. They gained money, energy and quality of life.

Place	What they built	Positive-sum gains they measured
Vienna, Austria	Waste-to-energy plant + city district heating	Heat for 60,000+ households from 250,000 t of waste; ~1.5 Mt CO_2 avoided per year; emissions ~90% below legal limits
Copenhagen, Denmark	CopenHill plant, owned by five municipalities	Heat and power for around 150,000 households; a landmark ski slope and recreation area on the roof
Kalundborg, Denmark	Industrial symbiosis since 1972 — 17 partners	~635,000 t CO_2 and ~€80 million saved every year; 4 million m^3 of water and 62,000 t of materials reused annually
Sweden (national)	Waste-to-energy feeding district heating nationwide	19.5 TWh recovered in 2023 — heats 1.47 M apartments, powers 940,000; ~50% of homes on district heating
Kawasaki, Japan	Eco-Town urban-industrial symbiosis since 1997	Incinerated waste down from 554,000 t (1990) to 356,000 t (2019) while population grew; ~13.8% lower CO_2e
European Union	Cross-industry symbiosis projects	Reviewed projects report cost savings of 20%–80%, mostly from reduced energy use, plus local job creation

Every one of these places started where Skopje is today. The technology is ready, the economics work, and the gains are shared. Skopje can be next.